

An operational reconstruction ladder for spacetime quantities from causal order

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Abstract. We study, in controlled $(1+1)$ -dimensional (and, for dimension, higher-dimensional) Minkowski models, which spacetime quantities can be operationally reconstructed from a causal (accessibility) order once a minimal, explicitly declared set of additional ingredients is supplied, and we organize the reconstructions as a ladder. Order statistics estimate dimension in flat Poisson-sprinkled Alexandrov intervals; supplying a global event density turns interval cardinality into a time-like proper-time estimate; an observer chain with clock labels yields radar time and unsigned radar distance; an orientation reference lifts the reflection degeneracy and recovers the Lorentz map between inertial protocols; overlapping charts yield an approximately consistent observer atlas; and supplied measure information makes volume reconstruction possible under the conformal ambiguity. Three negative results bound the ladder: causal order alone fixes no conformal scale, a single observer yields only unsigned distance, and finite signal speed alone does not produce Lorentzian structure; a Rindler horizon analogue confines radar reconstruction to the expected wedge. As a preregistered capstone we validate, on a $(3+1)$ -dimensional vacuum plane wave whose coordinate volume form and sprinkling law are exactly flat, that the conformal-class information order carries reaches an order-only statistic at finite density, at one deliberately strong operating point and as an existence result rather than a measurement of sensitivity. Separate preregistered stages extend both claim classes to a frozen Schwarzschild exterior patch (the paired claim confirmed, single-poset discrimination detected with incomplete separation), and a prediction-anchored stage finds the certified-membership count of a Poisson sprinkling concordant, within a frozen 2.5% band, with independently certified continuum 4-volumes on each rung of a four-rung mass ladder. The contribution is the explicit accounting of what each reconstruction requires and the negative results that bound it; we make no claim that spacetime is reducible to causal order.

Keywords: causal sets, spacetime reconstruction, radar coordinates, Poisson sprinkling, preregistered analysis

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1. Introduction

A conservative reading of the causal-set and order-first programs is that, for future- and past-distinguishing Lorentzian spacetimes, causal or chronological structure determines the conformal class. Supplying a volume element fixes the remaining conformal factor, up to diffeomorphism and unit conventions [1, 2, 3, 4, 5, 6]. This paper treats that

statement operationally and quantitatively. Rather than ask whether spacetime “is” a causal set, we ask: fix a causal (or null-accessibility) order in a known model; then, supplying an explicitly declared ingredient set, which spacetime quantities can a finite procedure actually reconstruct, with what error, and where does each reconstruction stop?

The value of posing it this way is an explicit *ledger*. Many individual reconstructions here are standard—proper time from longest chains [7], volume from interval cardinality [1], dimension from ordering fractions [8, 9], radar coordinates from an observer protocol. What is easy to lose, and what we make explicit, is exactly which extra structure each requires and what remains underdetermined without it. The ladder also isolates three instructive negative results: causal order alone does not fix conformal scale; a single observer yields only unsigned distance (a reflection degeneracy); and finite signal speed alone does not imply Lorentzian structure.

The contributions are: (i) a reconstruction ladder that indexes recoverable spacetime quantities by the minimal supplied ingredient set, with per-rung error behaviour in controlled models; (ii) grounded validations of dimension, proper time, radar decomposition, Lorentz-map and atlas consistency, and measure-dependent volume; (iii) three bounding negative results (conformal scale, reflection degeneracy, finite-speed lattice); (iv) a Rindler reconstruction-inaccessibility analogue; and (v) a preregistered (3+1)D capstone validation that a pure-Weyl deformation of the light cones—at identical coordinate volume form and sprinkling law—measurably moves the causal order at finite density (section 6), together with its type-D (Schwarzschild) extension, where separate preregistered stages confirm the paired claim and detect single-poset discrimination (section 6.7). All results are controlled validations, not evidence that geometry reduces to order.

2. The reconstruction ladder

We separate the *primitive* from *supplied ingredients* and index rungs by the minimal ingredient each requires.

The primitive is:

- **Causal / accessibility order.** A strict partial order on events (in the continuum models, Minkowski causal precedence; null-inclusive).

The supplied ingredient symbols (these form branches, not one linear hierarchy) are:

- **M—global event-density calibration.** The conversion between event count and physical volume in the declared sampling model.
- **O—observer chain with clock labels.** A timelike chain carrying ordered tick labels (an operational clock), defining a radar protocol.
- **R—oriented synchronized reference.** A second beacon chain with calibrated separation, fixing a spatial side.
- **A—observer atlas.** Multiple overlapping observer charts.
- **W—local measure profile / weights.** The physical volume element relative to the support sampling measure, normalized to unit mean over the declared

Table 1. Rungs of the reconstruction ladder: the minimal ingredient set, what it reconstructs, and what bounds it.

Rung	Ingredient	Reconstructed	Bounded by
R0	order only	flat-sprinkling dimension estimate; uncalibrated chain statistics	manifold/ model assumptions; no metric scale
R1	order + M	timelike proper time (interval cardinality $\rightarrow$ volume $\rightarrow \tau$)	needs a density
R2	order + O	radar time; unsigned radar distance	sign undetermined
R3	order + O + R	signed coordinates; Lorentz map between protocols	supplied orientation
R4	order + O + R + A	atlas transition maps (Poincaré); invariant agreement	oriented, calibrated charts
R5	order + M + W	volume under conformal ambiguity; coarse-graining stability	global and local measure supplied

support, so that all scale information belongs to M and W carries only position dependence; a constant profile is therefore $W = 1$ and supplies nothing beyond M.

Table 1 lists the rungs (minimal ingredient $\rightarrow$ what is reconstructed).

Three negative results bound the ladder from below (section 5): conformal scale is not fixed by order alone (motivating M and W); a single observer gives only unsigned distance (motivating R); and finite signal speed alone does not give Lorentzian structure. Figure 1 summarizes the ladder.

3. Methods: the verified foundation layer

Natural units, $c = 1$. The foundation modules were checked for numerical correctness against known results (sprinkling measure, Minkowski causal precedence, longest-chain and interval-cardinality normalizations, Myrheim–Meyer inversion, radar/Lorentz/Rindler formulas); this section states the conventions the results depend on. Figure 2 illustrates the primitive and two of the operational constructions.

- **Sprinkling.** Events are sampled uniformly with respect to the Minkowski volume in a causal diamond (rejection sampling on the ball-volume slice; equivalently uniform in null coordinates). The (1+1)D and D -dimensional diamonds and the (1+1)D forward cone are provided.
- **Causal order.** For events (t, x) , i precedes j iff $t_j - t_i > 0$ and $(t_j - t_i)^2 - |x_j - x_i|^2 \geq -\text{atol}$ (null-inclusive J^+ ; strict in time, so irreflexive).
- **Longest chain.** Topological-sort longest-path dynamic programming on the order; the (1+1)D proper-time normalization uses the Brightwell–Gregory constant

required supplied structure → reconstructed quantity	
R0	order only → flat-sprinkling dimension; raw chain statistics <i>bounded by: model assumptions; no metric scale</i>
R1	order + global density → timelike proper time <i>bounded by: needs a density</i>
R2	order + observer clock → radar time; unsigned distance <i>bounded by: sign undetermined</i>
R3	order + observer + orientation → signed coords; Lorentz map <i>bounded by: calibrated separation</i>
R4	order + observer + oriented atlas → Poincaré transition maps <i>bounded by: calibrated charts</i>
R5	order + global/local measure → volume; coarse-grain stability <i>bounded by: global/local measure supplied</i>

Figure 1. The reconstruction dependency ledger. Each row names a minimal ingredient set and the quantity it unlocks; the observer and measure rows are branches rather than one cumulative linear chain. The “bounded by” line records what that ingredient set still does not supply.

($L \sim \sqrt{2\rho}\tau$, i.e. $L \sim 2\sqrt{N}$ in a diamond of N events, consistent with section 4.2 and appendix A), with an endpoint-exclusive counting convention—the chain’s interior elements, two fewer than the count that includes both endpoints—and acknowledged finite-size corrections.

- **Interval cardinality.** The open Alexandrov interval count K between two events estimates volume $V = K/\rho$; in (1+1)D $V = \tau^2/2$, so $\tau_{\text{est}} = \sqrt{2K/\rho}$.
- **Myrheim–Meyer dimension.** The ordering fraction (related pairs over all pairs) is inverted against the known $f(d)$ curve to estimate spacetime dimension in flat Alexandrov intervals.
- **Radar coordinates.** For a supplied observer chain, τ_- and τ_+ are the latest preceding and earliest succeeding tick labels; radar time is their mean and radar distance their half-difference. A single chain gives unsigned distance; a second oriented chain supplies the sign. Lorentz maps between two inertial protocols are fit on their overlap [10].
- **Rindler.** An accelerated (Rindler) observer in flat spacetime, with analytic two-way radar ticks; the reconstructible region is the Rindler wedge, with the horizon appearing as a reconstruction-inaccessibility boundary [11].
- **Conformal / measure.** Positive conformal rescalings preserve causal order while changing volume and clock scale; volume reconstruction then requires sup-

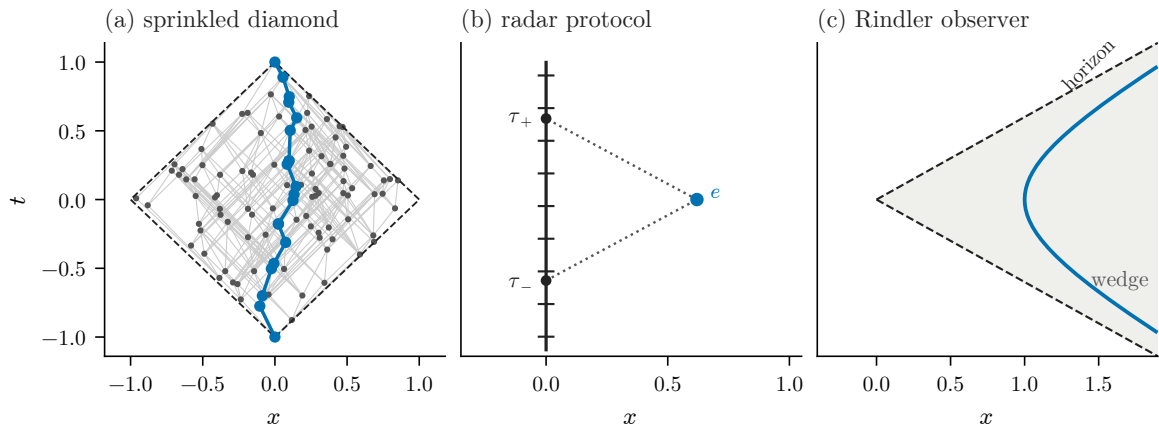


Figure 2. The primitive and two operational constructions (illustrations drawn with the foundation modules at a fixed seed; no measured quantity is displayed). (a) A Poisson sprinkling into a (1+1)D causal diamond, with its covering (Hasse) relations in gray and one longest chain between the diamond tips highlighted. (b) The radar protocol: an observer chain with tick labels, and the latest preceding (τ_-) and earliest succeeding (τ_+) ticks that bound a target event e . (c) An accelerated (Rindler) observer’s worldline; its two-way radar reconstruction is confined to the shaded wedge, whose boundary is the horizon.

plied measure information (implemented in `exp19` as local weights). In the tested uniform random-thinning protocol, reconstruction is stable after density rescaling.

4. Results

Each result is a controlled validation in a known model; we report the grounded number and its producing script.

4.1. $R0$ — order alone

Dimension. In the declared flat Poisson-sprinkling model, the Myrheim–Meyer order statistic estimates spacetime dimension: for true dimension 2 the estimate moves $1.95 \rightarrow 2.03 \rightarrow 2.01 \rightarrow 1.99$ at $N = 300, 600, 1200, 2400$; for dimension 3, $2.96 \rightarrow 2.95 \rightarrow 3.00 \rightarrow 3.00$; for dimension 4, $4.07 \rightarrow 3.97 \rightarrow 3.93 \rightarrow 3.97$. Endpoint RMSE is lower at $N = 2400$ than at $N = 300$, with non-monotonic finite-sample fluctuations for dimensions 3 and 4 (`exp10`; `outputs/data/dimension_reconstruction_summary.csv`).

Uncalibrated timelike statistic (longest chain). The longest chain follows the Brightwell–Gregory scaling $L \sim \sqrt{2\rho}\tau$, approached from below at finite density: at $\rho = 300$ the normalized length $L/(\sqrt{\rho}\tau)$ is 1.325 with endpoints included and 1.085 with endpoints removed, versus the asymptotic $\sqrt{2} \approx 1.414$. The proper-time estimator reads the endpoint-exclusive length, and its mean error there is -0.127 ; the same chains scored under the inclusive convention give -0.046 , the two separated by the exact offset $2/\sqrt{2\rho} = 0.082$. At this density the median chain carries 12.5 elements including its endpoints, so the two-element difference between the two conventions accounts for most of that residual: at these chain lengths it is a convention effect before it is a finite-size one (`exp09`; `outputs/data/longest_chain_calibration_summary.csv`). The approach toward the asymptotic value with N is shown by the chain estimator’s error falling

with N in section 4.2. Raw longest-chain length is order-theoretic. Interpreting it as proper time, including the normalization quoted here, belongs to R1 because it requires density and a dimension-specific asymptotic law; absolute scale is not fixed by order alone.

4.2. R1 — order plus measure: timelike proper time

With event density supplied, Alexandrov interval cardinality reconstructs timelike proper time between internal pairs, $\tau_{\text{est}} = \sqrt{2K/\rho}$. The volume-estimator relative RMSE falls from 0.273 at $N = 300$ to 0.105 at $N = 2400$. Its separately aggregated RMSE is below the chain estimator's (0.382 $\rightarrow$ 0.231) at every N , but the volume statistic uses 500 pairs per N whereas the chain statistic uses 125, so this is not a paired comparison, and the chain estimator additionally carries the endpoint-convention offset of section 4.1, so the comparison is not convention-free either (`exp07; outputs/data/timelike_pair_reconstruction_summary.csv`). In a fixed-interval sanity check, the interval-cardinality formula returns $\tau = 2.0$ exactly because the full sampling diamond uses $K = N$ and $\rho = N/V$; this is a normalization identity, not an independent accuracy test. The chain estimate converges toward it (0.303 at $N = 200$ to 0.0125 at $N = 2000$) (`exp03; outputs/data/timelike_reconstruction_summary.csv`). The observed errors are consistent with finite-sampling noise at the tested settings: the ratio of reconstruction RMSE to the predicted Poisson standard deviation is 0.93–1.07 across N (binomial 1.00–1.14). Smaller bias or other estimator errors are not excluded (`exp08; outputs/data/probe_pair_statistical_calibration_summary.csv`).

4.3. R2 — order plus observer: radar time and unsigned distance

Given an observer chain with clock labels, radar time and unsigned radar distance are reconstructed from causal accessibility; at $N = 300$ the error falls roughly by half per doubling of tick resolution—radar-time RMSE from 0.027 at 16 ticks to 0.0033 at 128 ticks, radar-distance RMSE from 0.071 to 0.0085, with the accessible fraction 1.0 throughout (`exp11; outputs/data/discrete_radar_reconstruction_summary.csv`). A single observer determines only unsigned distance (the reflection degeneracy of section 5).

4.4. R3 — plus orientation: signed coordinates and the Lorentz map

A second synchronized beacon chain with known separation supplies orientation, lifting the degeneracy to signed (1+1)D coordinates. The affine map recovered between two oriented inertial protocols approaches the Lorentz boost, with the fitted- β RMSE falling with tick density. Averaged over N as in figure 3, it falls from 0.00495 at 32 ticks to 0.000975 at 128 ticks for $\beta = 0.3$, and from 0.01345 to 0.00431 for $\beta = 0.6$ (`exp13; outputs/data/oriented_radar_lorentz_summary.csv`).

Figure 3 collects the convergence behaviour of the first four rungs: dimension recovery (a), timelike proper time (b), observer radar (c), and Lorentz-map recovery (d).

4.5. R4 — plus oriented atlas: transition-map consistency

With overlapping observer charts that retain the synchronized, calibrated orientation reference from R3, affine Lorentz/Poincaré transition maps fit on chart overlaps show approximately consistent composition and invariant agreement, improv-

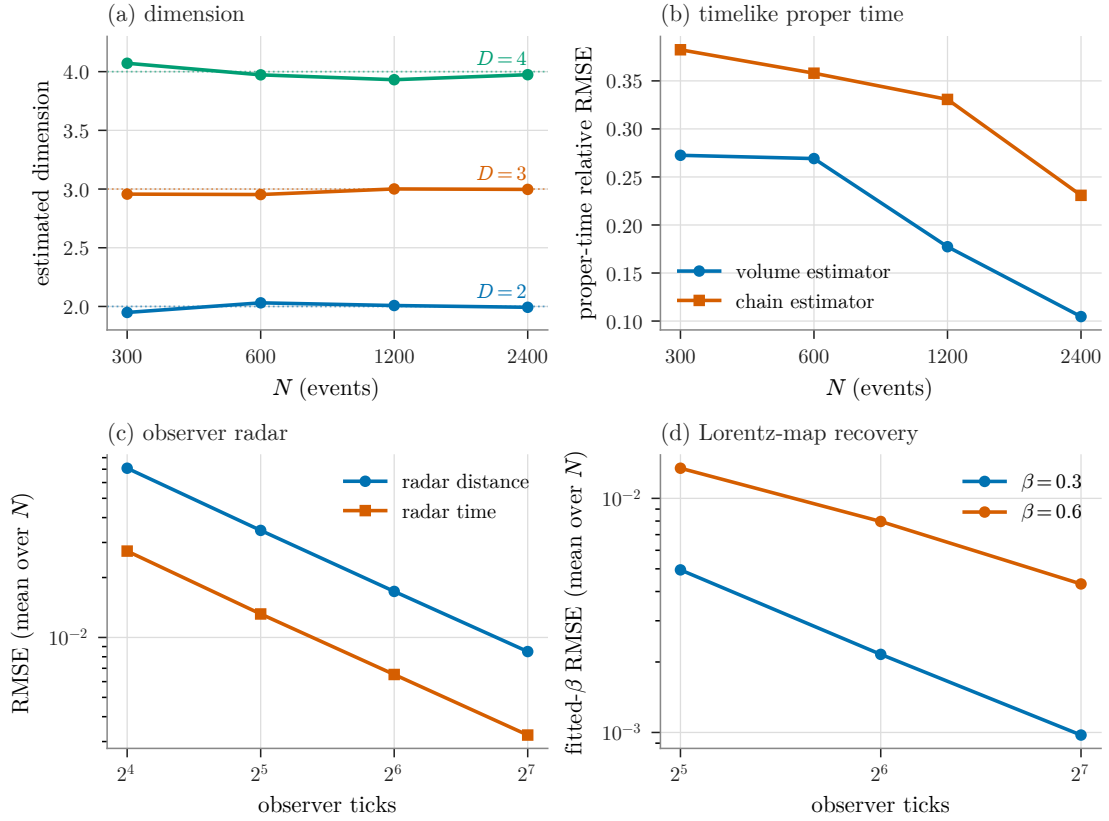


Figure 3. Reconstruction accuracy in controlled models. (a) Myrheim–Meyer dimension estimates lie near $D = 2, 3, 4$; endpoint RMSE is lower at $N = 2400$ than $N = 300$ despite non-monotonic finite-sample fluctuations (`exp10`). (b) Timelike proper-time relative RMSE falls with N , the interval-volume estimator below the separately aggregated longest-chain estimator (`exp07`). (c) Observer radar time and distance RMSE fall about by half per doubling of clock ticks (`exp11`). (d) Recovered Lorentz β RMSE falls with clock ticks for two boosts (`exp13`). Panels (c, d) are means over N ; axes are logarithmic where noted.

ing with tick density: the mean transition-map β error falls $0.0047 \rightarrow 0.0012$ and the invariant-interval RMSE $0.050 \rightarrow 0.010$ (ticks $32 \rightarrow 128$), while loop closure ($A \rightarrow B \rightarrow C$ versus direct $A \rightarrow C$) has a β -composition error of $0.0072 \rightarrow 0.0017$ (`exp14`; `outputs/data/observer_atlas_transition_summary.csv`, `outputs/data/observer_atlas_loop_summary.csv`). On exact analytic input the same transition/loop machinery recovers the maps to machine precision (β error $\sim 5.6 \times 10^{-17}$, RMSE $\sim 4 \times 10^{-17}$) (`exp15`; `outputs/data/exact_poincare_map_sanity.csv`).

4.6. *R5 — plus conformal measure: volume and coarse-graining*

Positive conformal rescalings preserve causal order while changing volume and clock scale: under constant (1.0/1.5/2.0) and sinusoidal rescalings the causal matrix is unchanged and the reconstructed dimension is identical (2.020), while the proper-time ratio tracks 1.0/1.5/2.0 and the volume ratio 1.0/2.25/4.0 (`exp18`; `outputs/data/conformal_order_ambiguity_summary.csv`). Two different things must therefore be supplied, and they fail differently when withheld: a scale (M, one number) and a shape (W, a field). The profile `exp19` tests is genuinely position-dependent—

$\Omega(t) = 1 + 0.3 \sin(\pi t/T)$ moves the local volume element by a factor of 3.45 across the diamond (0.49 to 1.69) while leaving every global summary near flat: the diamond volume moves by 2.7% (exp18’s sinusoidal row). A reconstruction holding only the global density therefore has almost no bias to correct (unweighted volume bias -0.0057 to -0.0028), but it carries a position-dependent error that no single rescaling removes: its relative RMSE stalls at $0.373 \rightarrow 0.298 \rightarrow 0.300$ across the tested N . Supplying the local weights removes the floor: the weighted relative RMSE falls $0.235 \rightarrow 0.176 \rightarrow 0.119$, tracking the flat-profile sampling floor ($0.234 \rightarrow 0.174 \rightarrow 0.118$) to within 1.3%, and the unweighted-to-weighted volume-RMSE ratio grows $2.0 \rightarrow 2.8 \rightarrow 4.3$ with N —the signature of a removed systematic rather than of reduced noise. What R5 demonstrates is that replacement of a non-vanishing error floor by flat-case, noise-limited error scaling—a consistency statement, not a bias correction (exp19; outputs/data/weighted_conformal_volume_summary.csv).

The constant-1.5 profile in the same table is a control, not the result: a constant weight rescales estimate and truth alike, so its weighted relative RMSE equals the flat profile’s at every tested N , and its unweighted bias (~ -0.29) is exactly the missing global factor—a normalization identity in the sense of section 4.2, whose repair is R1’s ingredient M, not W. The analytic volume/proper-time formulas are verified to $\sim 1 \times 10^{-7}$ (exp20; outputs/data/conformal_volume_exact_sanity.csv). Under random thinning, density-rescaled reconstruction is stable (volume RMSE $0.018 \rightarrow 0.009$, dimension steady ~ 2.0) while the uncorrected estimate blows up to RMSE 0.270 (bias -0.183) at 25% retention; the thinning is uniform, so its keep probability is one global scalar and this is a robustness statement about M—the number-volume correspondence survives uniform decimation once the density is re-declared—not a second W-level result (exp23; outputs/data/thinning_coarse_graining_summary.csv). Figure 4 shows the measure dependence directly.

4.7. Horizon analogue — Rindler reconstruction-inaccessibility

For an accelerated (Rindler) observer in flat spacetime, two-way radar reconstruction is confined to the Rindler wedge, and nothing leaks out of it: in every configuration the classification records zero false positives and zero false negatives, precision = recall = 1.0. That scoring is against the events finite clock coverage makes accessible, not against the ideal wedge; measured against the ideal wedge the same reconstruction is incomplete, recovering a fraction 0.74 to 0.84 of it, because part of the wedge lies beyond the observer’s tick range. The wedge itself covers about a quarter of events (accessible fraction ~ 0.25), and the in-wedge radar-time RMSE falls with tick resolution (e.g. $0.0098 \rightarrow 0.0023$) (exp16; outputs/data/rindler_horizon_reconstruction_summary.csv). Comparing observers directly, all events are accessible to the inertial observer while only ~ 0.25 (ideal wedge) / ~ 0.21 (finite clock coverage) are accessible to the Rindler observer, and no event is Rindler-accessible but inertial-inaccessible—a strict subset (exp17; outputs/data/inertial_vs_rindler_accessibility.csv). This is a controlled flat-spacetime horizon analogue, not a black-hole simulation.

5. Negative results that bound the ladder

- **Conformal scale is not fixed by order alone.** Positive conformal rescalings leave the causal order invariant while changing physical volume and clock scale

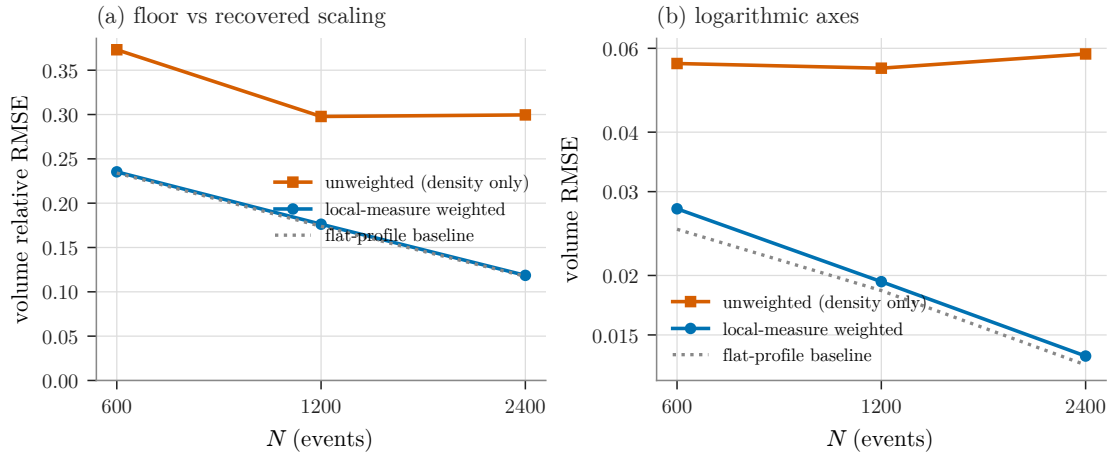


Figure 4. R5: what the local measure buys. For the position-dependent profile, the unweighted estimate’s relative error stalls at a floor that no global recalibration removes, while the weighted estimate falls with N onto the flat-profile sampling floor (left); on logarithmic axes the two carry visibly different slopes (right). The constant-profile arm is not plotted: a constant weight rescales estimate and truth alike, so its weighted curve is the flat curve exactly—that arm is a normalization identity, and its repair belongs to R1. Equivalent measure information can conceptually be encoded in the sampling law rather than as post-hoc weights; that alternative is not part of this figure’s evidence package. Order alone fixes no absolute scale, and no scale alone fixes a profile.

(section 4.6). Absolute scale therefore requires a supplied measure (rung M); R0 therefore supplies uncalibrated order statistics, not a proper-time scale.

- **A single observer gives only unsigned distance.** One chain’s radar distance is $|x|$: two targets at $x = +0.1$ and $x = -0.1$ return the identical single-observer distance 0.1, while the two-chain oriented protocol recovers the signed positions $+0.1$ and -0.1 . Signed coordinates therefore require an orientation reference (rung R) (`exp12; outputs/data/single_observer_reflection_degeneracy.csv`).
- **Finite signal speed alone does not give Lorentzian structure.** After calibrating density at the final time, a regular finite-speed lattice shares the continuum model’s leading quadratic count growth, while finite- t counts differ ($t = 5$: 21 vs 14.75; $t = 30$: 496 vs 496). Yet its edges lie only along the two lightcone diagonals (465 each), so it has a discrete symmetry, not the continuous Lorentz symmetry of a sprinkled causal set (`exp05; outputs/data/finite_speed_lattice_growth.csv`). Finite speed is necessary but not sufficient; statistical Lorentz compatibility is the additional structure.

An exploratory spacelike-distance proxy (common-past / common-future / enclosing-interval counts) is reported as boundary-dependent and *not* a validated spacelike estimator: the counts vary strongly with the sampling region rather than tracking spacelike distance cleanly (`exp06; outputs/data/spacelike_distance_proxy_summary.csv`).

6. Capstone validation: finite-density detectability of the conformal class

The ladder reconstructs quantities from supplied ingredients inside flat models; section 5 bounds it from below. This section caps it from the other side, with the converse of the first negative result. Section 4.6 showed that *within* a conformal class, order carries no scale: positive rescalings leave the causal matrix unchanged. By the theorems that motivate the program, order determines the conformal class itself—so *across* conformal classes order must differ. Whether that difference survives finite density, for a curvature channel that touches nothing but the order, is the question this capstone answers affirmatively, as a preregistered experiment (P14 in the program’s internal numbering; within this program, the first finite-density crossing of the conformal-flatness ceiling, via a 4D type N pure-Weyl construction).

6.1. The (1+1)D conformal ceiling

Every (1+1)D metric is conformally flat, so in the (1+1)D stages of this program (and its curvature-focused companions P12–P13) curvature could reach the order only through the volume-coefficient channel: those experiments tested curvature entering through the measure/counting channel, and the channel in which order directly carries a conformal-class difference could not be tested there at all. This was an intended dimensional limit of the models, not a confounded experiment: Weyl curvature—the conformal-class content of curvature—is identically zero below four spacetime dimensions, so a direct test needs a 4D construction built so that the measure channel is exactly frozen while the conformal class moves.

6.2. Pure-Weyl control construction

The construction is a vacuum plane wave in Brinkmann coordinates with profile $A(u)(x^2 - y^2)$ on a constant- A slab: Ricci-flat, Petrov type N, with $\det g = -1$ identically. Because $\det g = -1$, the coordinate volume form and the Poisson sprinkling law are *identical* to flat spacetime. That identity supports the strongest form of pairing, and *C1 uses it*: C1’s control reading is the *same sprinkled point set* read with $A = 0$, so for C1 any difference between the two readings is a functional of the relation change alone—it cannot be a sampling artifact, an intensity mismatch, or a calibration residual. C2 deliberately does *not* pair: its two arms are independent, unpaired sprinkling streams drawn under the identical sprinkling law, with finite-sample variation controlled by its frozen interval rules rather than removed by pairing—because its claim is about telling one ensemble’s posets from the other’s, not about a within-point-set counterfactual (section 6.3). What is *not* identical between geometries is the causal-diamond volume: the light-cone boundary tilts, and at equal proper time the curved-arm diamond volume sits 0.4% ($wT = 1$) to 7.3% ($wT = 2$) above flat—a deterministic design check pins both numbers, against the closed form $V_A/V_0 = 1 + (wT)^4/252 + O((wT)^8)$. Those percentages describe a small diamond on the axis, and the frozen operating point is neither small nor on-axis: the probe chain selected **aniso-a1.0**, a slab spanning $1/\pi = 0.318$ of the way to the first conjugate point, on a box elongated three to one along the focusing transverse axis. The order statistic therefore reads mostly off-axis pairs in the direction where the cones open, which is why its response is a factor of three rather than a few percent. This is a deliberately strong operating point, not a weak-curvature probe. “Same measure” here means the volume form and sprinkling law only, never interval

Table 2. Preregistered capstone verdicts. Both claims confirmed; the stage is POSITIVE.

Claim	Verdict	Result (95% CI)
C1 paired ensemble mean, $n = 3000$	confirmed	mean 0.0502929 [0.0501046, 0.0504812]; lower end $140\times$ the margin 3.579×10^{-4}
C2 classifier replication, $n = 4800/\text{arm}$	confirmed	separation $s = 11.198$ [11.035, 11.362]; AUC = 1.0 [0.999232, 1.0]; balanced accuracy = 1.0 [0.986, 1.014]

volumes; conflating the two was the first design draft’s central error, caught in review, and the distinction is load-bearing.

6.3. Finite-density detection design

An exploratory probe chain (four probes and a confirmation stage, run on seed streams disjoint from everything that follows) selected the operating point—slab (1.0, 1.0, 2.0, 6.0), $w = 1.0$, expected 300 events per sprinkling—and the primary statistic, the global relation fraction (the one candidate computable from a single poset alone). The preregistered stage then froze two claims with deliberately different boundaries, never merged into one sentence:

- **C1 (paired ensemble mean).** The estimand is $\theta_\Delta = E[f_A - f_0]$, the ensemble mean of the *net* change of the relation fraction over paired readings of the same points—not any single point set’s displacement, and not the gained-plus-lost total motion. Confirmation requires the 95% Student- t interval of the mean paired difference ($n = 3000$ sprinklings) to sit entirely above a frozen margin $\varepsilon_\Delta = 3.579 \times 10^{-4}$, an operational margin anchored to the probe chain’s confirmed block.
- **C2 (single-poset classifier, independent replication).** A frozen classifier reading only the single-poset relation fraction must separate curved from flat ensembles ($n = 4800$ per unpaired arm) under three frozen interval rules (standardized separation, AUC with an exact boundary bound at complete separation, balanced accuracy), each with its own equivalence band. This claim replicates the probe chain’s confirmation stage on fresh seeds; its failure would have been reported as a conflicting replication, never a retroactive cancellation.

The stage verdict POSITIVE requires both claims confirmed, certified as a joint event before execution (4000/4000 joint-effect replicates; the null branches certify each claim’s equivalence power separately at an exact Clopper–Pearson [12] 95% lower bound of at least 0.90; a joint equivalence verdict is deliberately not defined, because its joint null rate sits below that floor at the frozen sizes).

6.4. Preregistered result

The campaign ran once, from a clean checkout of the freeze commit, on frozen seeds, and recorded the executing commit in the artifact. Both claims confirmed; the stage is POSITIVE (table 2). Figure 5 shows both claims.

The size of the effect is worth stating in the statistic’s own units, which the margin ratio hides: averaged over the paired readings the relation fraction runs 0.023830 flat

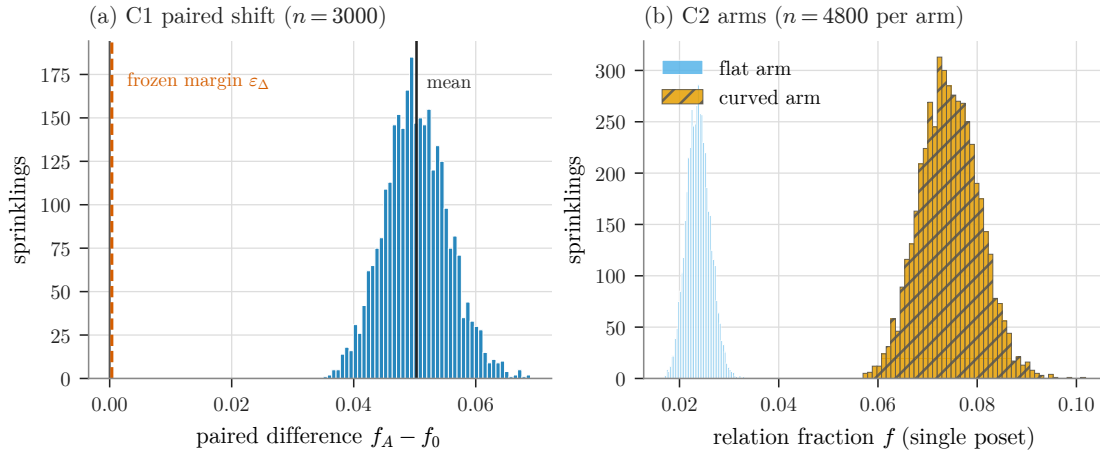


Figure 5. (a) The distribution of the paired per-sprinkling relation-fraction differences $f_A - f_0$ ($n = 3000$), with mean 0.0502929 (95% CI [0.0501046, 0.0504812]); the CI lower end is $140 \times$ the frozen margin $\varepsilon_\Delta = 3.579 \times 10^{-4}$ (marked); (b) the single-poset relation-fraction samples of the two unpaired arms ($n = 4800$ per arm), which are completely separated (the frozen exact-bound AUC interval is [0.999232, 1.0]). Both panels are drawn from the committed artifact (`docs/prereg/p14_prereg_results.json`); no value is typed in.

against 0.074123 curved, a factor of 3.11. The two arms’ relation-fraction samples are completely separated (the minimum curved-arm value exceeds the maximum flat-arm value in 4800 draws per arm); the AUC interval at complete separation is the frozen exact bound, not the degenerate Wald interval. The balanced-accuracy interval got no such treatment: its frozen construction is an unclipped Wald interval on a worst-case standard error, so at $BA = 1.0$ it reaches 1.014, outside the range the quantity can take. That is what the preregistration licenses and it is printed as it stands; no verdict depends on it, because the gate reads only the lower end. For the reader’s orientation—not as a re-scored gate—the construction the later S5 stage uses for the same quantity, a Clopper–Pearson interval per arm combined across the two, gives [0.998176, 1.0] on the same test halves. Every relation census in both claims recorded zero ambiguous and zero escalated pairs. The two sentences the stage licenses were fixed in the preregistration before execution and are recorded in the artifact; in translation they are C1, *the paired ensemble mean shift exceeds ε_Δ* , and C2, *the probe chain’s separation is independently reproduced* (`docs/prereg/p14_prereg_results.json`; appendix B records the language of the frozen originals and what holds these renderings to them).

6.5. What this establishes

At a fixed box, density, and profile, an order-only statistic separates flat from curved ensembles: the conformal-class information that order carries by theorem is *present in the order at finite density*, in the one construction where the sprinkling measure is exactly frozen. The claim is existence, not sensitivity. No density or amplitude sweep was run, so nothing here locates the point at which the separation would fail, and the operating point was chosen to be strong (section 6.2) rather than marginal. Together with section 4.6 this closes the conformal story in both directions—within a class, order is blind to scale; across classes, at least at this operating point, order visibly moves.

6.6. What the plane-wave result alone does not establish

It says nothing about sensitivity. The frozen operating point was chosen to be strong, and two things follow that the stage cannot answer: no density or amplitude sweep was run, so the density at which the separation fails is unmeasured; and the slab is elongated three to one along the focusing transverse axis, so whether the effect survives at an isotropic box is untested. Beyond that, it is not a general-Weyl discriminator (the plane wave is Petrov type N and its exact volume form is unrepresentative); it is not Weyl-tensor recovery (detection and recovery are different claims; no reconstructed quantity is produced, which is why this is a capstone boundary experiment and not a new ladder rung); it does not establish box- or density-independence (the licensed sentence is bound to the frozen operating point); and, by itself, it does not open the Schwarzschild generalization. Both claim classes are carried there by the separate pre-registered extensions of section 6.7—the C1 paired claim confirmed, C2 single-poset discrimination detected with incomplete separation; the diamond-volume oracle is now certified, and its direct-MC instrument audit is complete as an auxiliary result (section 9, appendix B), and the prediction-anchored Poisson-count stage is executed across a four-rung mass ladder, CONCORDANT at every rung (section 6.8). A separate cost measurement (S1) prices the causal-predicate component—about 0.77 ms per pair on the tested solver, patch, and tolerance, roughly $360\times$ the plane-wave predicate (`docs/prereg/p14_s1_cost.json`).

6.7. Type-D extension: preregistered C1-paired confirmation and C2-unpaired detection in Schwarzschild

The paired half of the capstone design transfers to the Schwarzschild exterior on a measure identity: in Schwarzschild coordinates the volume element is $\sqrt{-g} = r^2 \sin \theta$ —independent of the mass and equal to the flat spherical element—so one sprinkle of a coordinate patch serves both geometries, exactly as $\det g = -1$ served the plane wave. What the identity controls is the sampling measure; individual diamond volumes differ under the two causal structures, and that difference is part of the signal (the section 6.2 distinction, unchanged). The frozen patch is the S1 solver domain ($M = 1$, exterior shell $r \in [10, 20]$, polar cap, coordinate-time extent 40), with $N \sim \text{Poisson}(300)$ events per reading and the S1 predicate at tolerance 1×10^{-8} with escalation to 1×10^{-10} .

An exploration stage (S3, seed-ledger disciplined, preserved with its raw per-reading arrays) measured the paired shift and sized a preregistered confirmation (S4). The frozen rule (`docs/prereg/p14_s4_schwarzschild_c1.md`) fixes two gates evaluated on identified intervals with strict comparisons: a primary detection gate (identified CI95 above the frozen threshold $\varepsilon_{\text{det}} = 0.0036$, about 10% of the exploration anchor, in the frozen negative direction) and a secondary replication gate (the independent two-sample Welch CI95 of the S4–S3 difference inside $\pm\varepsilon_{\text{rep}} = \pm 0.0012$, one exploration reading-SD); a verified inclusion makes the joint verdict numerically equivalent to the replication gate at these margins. Branch powers were certified before the freeze by resampling both blocks from a conservatively scaled empirical distribution (20000/20000 on every branch, exact Clopper–Pearson 95% lower bound 0.9998), with a constructed negative-control battery proving each gate can fail. The campaign ran once, from the exact freeze checkout, with an 8-file content-addressed manifest verified at entry and at exit; the frozen gates and results are in table 3.

Table 3. S4 frozen gates and results.

Gate	Frozen condition	Result	Verdict
A: C1 detection (primary)	identified CI95 top < -0.0036	CI95 $[-0.036211, -0.035953]$	pass ($10\times$ margin)
B: replication (secondary)	Welch CI95 of S4–S3 inside ± 0.0012	$[-0.000183, +0.000194]$	REPLICATED

Table 4. S5 frozen gates and results.

Gate	Frozen condition	Result	Verdict
Primary: AUC (DeLong)	CI95 lower > 0.60	AUC 0.9734, CI95 [0.9630, 0.9837]	DETECTED
Secondary: out-of-sample BA	joint CI95 lower > 0.60	BA 0.9033, CI95 [0.8356, 0.9500]	pass

Stage verdict: CONFIRMED. The census recorded zero ambiguous pairs and one escalated pair, resolved at the tighter tolerance. The two sentences the stage licenses, in translation (`docs/prereg/p14_s4_results.json`), are *on the frozen Schwarzschild coordinates and domain, the paired ensemble mean shift exceeds the frozen detection threshold in the frozen direction—a finite-density causal census detects the light-cone deformation of type-D vacuum curvature at C1 grade; a program-internal statement, and the independent two-sample difference between the S4 and S3 blocks lies inside the replication band: the exploration effect is quantitatively reproduced.*

A second, separate preregistered stage (S5) carries the single-poset claim class. Guided by a committed stored-data feasibility audit (`docs/prereg/p14_c2_feasibility_audit.json`, exploratory, zero additional solver calls, conservatively anchored), its rule (`docs/prereg/p14_s5_schwarzschild_c2.md`) declares—before sizing and independently of the confirmation data—AUC 0.60 as the minimum practically useful single-poset discrimination. Two *independent* arms (fresh Schwarzschild and fresh flat sprinkles; 300 readings each by a pre-declared minimum- n rule with directly certified branch powers) score each reading by its global relation fraction; the primary gate is an unclipped DeLong [13] AUC CI95 with a four-way outcome (detected / equivalent-at-margin / direction-reversed / inconclusive, strict comparisons, identified-agreement across the curved bound series), and a secondary out-of-sample balanced-accuracy gate uses a deterministic Clopper–Pearson–Bonferroni interval with a train-half-only threshold. The campaign ran once from its own exact freeze checkout; the frozen gates and results are in table 4.

Zero ambiguous and zero escalated pairs; both identified bound series agree. The two sentences this stage licenses, in translation (`docs/prereg/p14_s5_results.json`), are *on the frozen Schwarzschild domain and density, the global relation fraction of a single causal set carries information that discriminates flat from Schwarzschild ensembles above chance—AUC CI95 lower bound above 0.60; a program-internal statement, and the joint 95% lower bound of out-of-sample balanced accuracy exceeds 0.60: out-of-training discrimination holds; a secondary verdict.*

What this extends, and at what grade: on the same frozen Schwarzschild domain, the two claim classes of the plane-wave capstone are now each supported by a *separate*

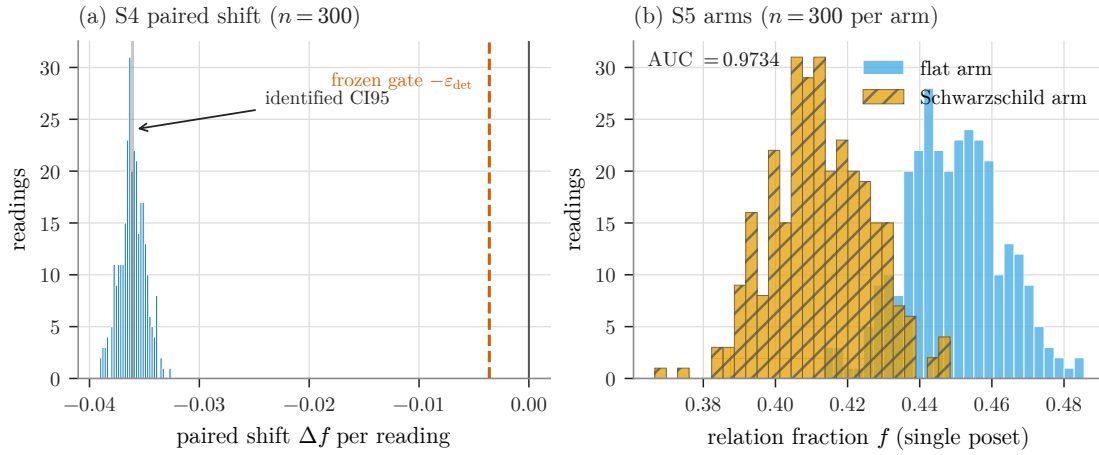


Figure 6. The Schwarzschild extension. (a) The per-reading paired shift under the frozen Schwarzschild domain: the identified CI95 $[-0.036211, -0.035953]$ lies an order of magnitude beyond the frozen detection threshold $\varepsilon_{\text{det}} = 0.0036$ (frozen negative direction). (b) The per-reading global relation fraction of the two independent arms, discriminating with AUC 0.9734 (DeLong CI95 $[0.9630, 0.9837]$)—strong but, unlike the plane-wave C2, incomplete separation. Drawn from the committed artifacts (`docs/prereg/p14_s4_results.json`, `docs/prereg/p14_s5_results.json`); no value is typed in.

preregistered stage (figure 6)—the C1 paired claim CONFIRMED (S4), C2 single-poset discrimination DETECTED (S5). There was no joint primary verdict. The two stages also differ in what they can resolve, and the difference is one of design rather than of curvature type. C1 reads the same sprinkled points twice, so its estimator sees only the relation change: the paired readings correlate at 0.9965 and pairing removes 99.3% of the spread, leaving a paired SD of 0.001135 against a detection threshold an order of magnitude away. What pairing removes is the shared point configuration itself, not the event count—that count correlates with the paired difference at -0.018 and is not the nuisance that matters. C2 reads two independent arms and can remove none of it: its pooled SD of 0.013428 is an order of magnitude larger, and against that the same geometric effect is worth 2.73 standard deviations rather than 31.8. Much of the gap is simply where each construction sits—the flat-arm relation fraction averages 0.0238 in the plane wave against 0.4491 here, and a fraction near one half is near its most variable. That, not type D versus type N, is why the plane wave’s C2 separated completely and this one does not ($\text{AUC} \approx 0.973$; no completeness gate was preregistered). A single-poset statistic that recovers what pairing sees is a natural next preregistration; recovering it after the fact would forfeit the freeze. The secondary BA verdict neither strengthens nor combines with the primary. The extension establishes no mass-generality (a single frozen M) and uses no diamond-volume oracle (margins operationally anchored or independently declared)—the mass ladder and the certified-volume anchor belong to the separate stage of section 6.8, which does not promote these two verdicts. Execution provenance for both stages, including the executed-freeze manifest snapshots—and the capstone’s own freeze ordering, mechanical gates, and commit-ancestry contract—is in appendix B.

6.8. Prediction-anchored Poisson count, and its mass generalization

Every verdict so far is anchored operationally: a margin taken from an exploration block, or a threshold declared before sizing. This stage is different. The diamond-volume oracle certifies an interval for the continuum 4-volume V of a fixed causal diamond, by directed-rounding arithmetic and before any count data exists; a Poisson sprinkling then supplies a count of elements certified to lie inside that same diamond. The question is whether the operational instrument realizes the certified prediction. Nothing in sections 6–6.7 is used to answer it, and the answer cannot be tuned after the fact: the endpoints, the intensity, the tolerance and the decision rule are all frozen in the rung’s preregistration before its seed is drawn.

The instrument. Points are sprinkled at a frozen intensity A into the certified box, and each point’s membership is decided by the certified causal predicate on both legs. A point the predicate cannot decide at the frozen tolerance is *not guessed*: it is counted as ambiguous (U_{amb}) and enters the decision only through the conservative end of the interval. Each rung first runs a fixed- n ambiguity pilot (exact Clopper–Pearson plus an exact Poisson tail bound at $\alpha = 0.01$, tail budget 1×10^{-3}) that must certify $U_{\text{max}} \leq 30$ before the count campaign is sized; the campaign’s membership test short-circuits on a decided first leg, so a point the campaign calls ambiguous is necessarily one the pilot’s test would also call ambiguous, and the pilot’s bound transfers *a fortiori*.

The frozen rule. From the realized $(K_{\text{certain}}, U_{\text{amb}})$ the rule builds a deliberately conservative *outer* count interval—the exact Garwood [14] lower limit at K_{certain} , the upper limit at $K_{\text{certain}} + U_{\text{amb}}$, each tail at level 0.025, so the ambiguous points can only widen the interval outward and never move it toward agreement. Rescaled by the intensity this is a volume interval $C = [L/A, U/A]$, and the quantity gated is the *identified discrepancy* against the certified enclosure,

$$D = [C_{\text{lo}} - V_{\text{hi}}, C_{\text{hi}} - V_{\text{lo}}], \quad \text{required to lie in } [-B, +B], \quad (1)$$

$$B = \tau V_{\text{ref}}, \quad \tau = 2.5\%,$$

with three ways out and no fourth: contained gives CONCORDANT, disjoint gives DISCORDANT, and anything else gives INCONCLUSIVE, published as it falls. The arithmetic is 96-bit end to end, and contract tests prove at the integer boundaries that the frozen acceptance window is exactly the set of counts the decision function calls CONCORDANT, so “the sizing” and “the verdict” cannot drift apart.

The ladder. One rung would establish the gate at one geometry. To ask whether the agreement is a property of the instrument rather than of a lucky configuration, the same gate is carried across a preregistered mass ladder. The certified shell [10, 20] and the anchors (12, 18) stay fixed in *absolute* coordinates, so no rung is an isometric copy of another: the pre-frozen dimensionless indicator is the compactness $\mu = 2M/r_c$ at the anchor midpoint $r_c = 15$, and the executed ladder spans a factor of three in it. The mass is not the only thing that moves. The diamond’s coordinate-time window is retuned per rung by the frozen rule $dt(M) = 8.5 T_{\text{min}}(M)/T_{\text{min}}(1)$, so that each rung opens the same multiple of its own minimal flight time; the whole diamond therefore scales with that flight time. The consequence is worth stating plainly, because it bounds what the ladder can show: the certified volumes track $(dt/dt_1)^2$ to 0.007%, 0.021% and 0.144% at $M = 1.4, 1.8$ and 3.0 , so their mass-specific content sits one to two orders of magnitude below the $\tau = 2.5\%$ band the gate applies. The deep end is not a free choice

Table 5. The four executed mass-ladder rungs: certified volume interval V , pilot ambiguity outcome, sprinkled N , certain and ambiguous counts, outer count interval C , identified discrepancy D , band B , and verdict.

μ	M	certified V	pilot	N	K_{certain}	U_{amb}	C	D	B	verdict
0.1333	1.0	[56.492959, 57.060667]	$k = 0$	26,831,117	53,285	0	[56.205871, 57.169553]	[-0.854796, +0.676594]	1.419420	CONCORDANT
0.1867	1.4	[64.336198, 64.982744]	$k = 1$	15,057,284	53,829	1	[64.307488, 65.405648]	[-0.675256, +1.069450]	1.616487	CONCORDANT
0.2400	1.8	[73.968637, 74.711920]	$k = 0$	9,847,124	53,513	0	[73.695211, 74.956037]	[-1.016709, +0.987400]	1.858507	CONCORDANT
0.4000	3.0	[121.225021, 122.443280]	$k = 0$	4,605,671	53,606	0	[120.802632, 122.867592]	[-1.640649, +1.642571]	3.045854	CONCORDANT

either: every mass-generic lemma certifies exactly on $M \in [0.92, 3.33]$ —bounded above by the photon sphere $3M$ reaching the shell floor at $M = 10/3$ —and the deepest rung $M = 3.0$ keeps a certified margin of a tenth of the shell floor on that binding condition, its inner anchor at $r = 4M$ between the photon sphere and the ISCO. The patch-level lemmas are mass-generic only under stated conditions (horizon below the shell, $K < 0$, w monotone, $Q > 0$, the L2a patch bound, four L4 margins, an L5 winding cross-check), and each is re-certified per rung as an interval comparison—nothing is inherited silently from $M = 1$.

Four rungs were executed, each from its own exact freeze checkout, with its own seed, once (table 5).

The certified volume grows monotonically with the compactness (57 to 65 to 74 to 122), which at this resolution is the retuned time window growing rather than an independent geometric signal, and the realized discrepancy stays well inside the band at every rung (figure 7). The $M = 1.4$ pilot returned the program’s first non-zero ambiguous count ($k = 1$); it was absorbed exactly as the sizing had provisioned, and it is visible in that rung’s asymmetric D , which is the conservative widening working as designed rather than a defect.

What this establishes. On each preregistered rung independently, an operational finite-density instrument realized that rung’s independently certified continuum volume to within $\tau = 2.5\%$, under a rule fixed before the data existed. This is the first place in the program where a measurement is confronted with a certified *prediction* rather than with an operational anchor, and the confrontation is repeated at four masses rather than one—the deepest with its inner anchor between the photon sphere and the ISCO. What the repetition buys is not four independent curvature probes: as noted above, the certified volumes differ from a flight-time rescaling by far less than the band, so the volume comparison itself is near-degenerate across rungs. What it does buy is that every patch-level lemma is re-certified and the causal predicate re-run at each mass, none of it inherited from $M = 1$.

What it does not establish. The rungs are separate preregistered stages: each verdict stands alone, there is no joint primary verdict, and no interpolation, extrapolation, or composite cross-rung statistic is claimed—in particular nothing is claimed at compactness between or beyond the four tested values. Agreement of a count with a volume is not a test of causal-set theory, and it is not evidence that spacetime is

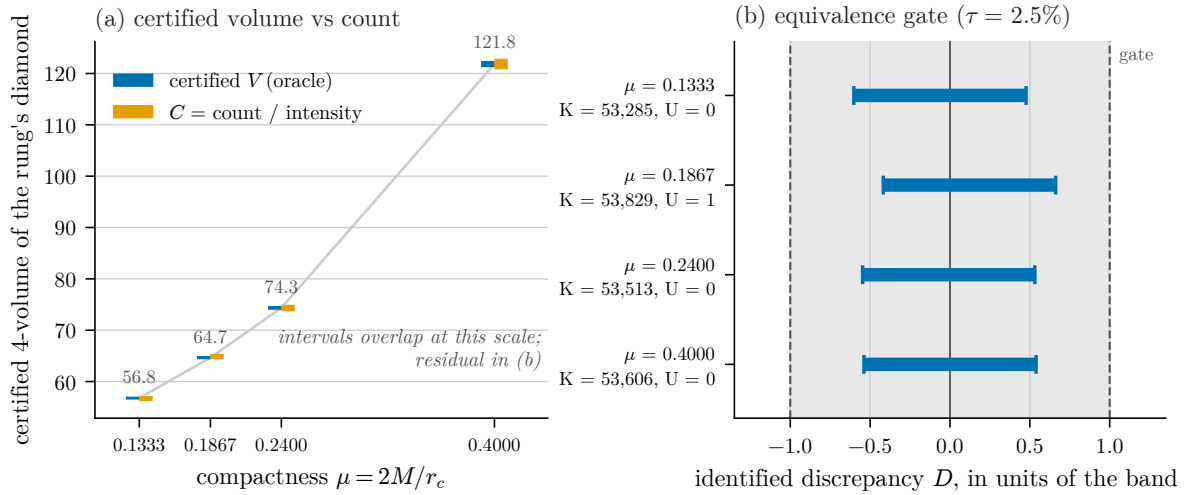


Figure 7. The executed ladder. (a) At each preregistered compactness, the certified continuum volume (blue) and the volume implied by the sprinkled count (orange). The intervals overlap at plot scale across a threefold span in μ —that overlap is the result. (b) The residual made visible: each rung’s identified discrepancy D divided by that rung’s own band $B = \tau V_{\text{ref}}$, so the rungs are comparable despite different bands; the shaded region is the gate. Every rung is contained, with roughly half the band to spare, and the $\mu = 0.1867$ rung’s asymmetry is its single ambiguous point widening the interval outward as the rule provides. Both panels are generated from the same artifacts the contract tests re-derive (`figures/make_ladder_figure.py`); no value is typed in.

discrete; it is a statement that this instrument, at these operating points, reproduces the continuum measure it was pointed at. It upgrades neither the C1/C2 verdicts of sections 6–6.7 nor the auxiliary O4b instrument audit, none of which use it. And the result is bounded by its construction exactly as the capstone is: one fixed diamond, one fixed density per rung, a Schwarzschild exterior shell, and a tolerance chosen to be feasible rather than to be sharp.

7. Discussion

The ladder is a branched accounting device, not a single cumulative chain. Across its dependency rows it recovers a substantial fraction of operational Lorentzian geometry—dimension, timelike duration, radar decomposition, Lorentz and Poincaré consistency, volume—from a causal order plus short, explicit ingredient sets. It is equally a list of what each reconstruction *costs*: absolute scale costs a measure; a signed spatial coordinate costs an orientation; a metric-not-merely-conformal statement costs a volume element; Lorentzian structure is not bought by finite speed alone. Stating both directions together is the point: it turns “causal structure fixes the conformal class, and a volume element fixes the remaining conformal factor” from a slogan into a per-quantity operational ledger with measured error behaviour.

This framing also clarifies what the program does *not* show and sets up the open question. Everything here is reconstruction inside known models: the geometry is put in (by sprinkling from Minkowski or by a supplied protocol) and recovered. It does not address whether geometry could *emerge* from an order that was not built from

a geometry—the representability question—which requires a validated discriminator and is taken up separately. The section 6 capstone sharpens the boundary from the other side: the one piece of geometry that order itself carries—the conformal class—is not merely present by theorem but legible to an order-only statistic at finite density, exactly where the measure channel is frozen and only the light cones move; how far that legibility extends before it fails is unmeasured.

8. Claim boundary

We claim, as controlled validations in known (1+1)D (and higher-D for dimension) models: dimension is estimated from order statistics in flat sprinklings; timelike proper time is recoverable from interval cardinality once a density is supplied, with finite-sampling-consistent error; radar time and unsigned distance are recoverable from an observer protocol, signed coordinates and the Lorentz map with an orientation reference, and atlas transition maps with overlapping oriented, calibrated charts; volume is recoverable with supplied global density and local measure information and is stable under density-rescaled coarse-graining; and a Rindler wedge is the reconstructible region for an accelerated observer. As the preregistered capstone (section 6), at the frozen (3+1)D operating point: the paired ensemble mean of the relation-fraction change under a pure-Weyl deformation exceeds its frozen margin (C1), and a frozen single-poset classifier separates curved from flat ensembles, independently replicating the probe chain’s confirmation (C2).

We do not claim: that spacetime is reducible to, or emerges from, causal order; that causal order alone yields absolute scale, the conformal factor, signed coordinates, or a unique atlas; that finite signal speed implies relativity; or that any of these finite validations establish a physical theory. Reconstructing a geometry that was put into the model is not deriving geometry from order. For the capstone we additionally do not claim: a general-Weyl discriminator, Weyl-tensor recovery, or box- or density-independence of the separation. The Schwarzschild path carries its own preregistered verdicts (section 6.7: C1 confirmed and C2 detected with incomplete separation, in separate stages with no joint primary verdict) and its own non-claims; the diamond-volume oracle is certified and instrument-audited there as an auxiliary result (section 9), but no section 6.7 verdict uses it. We claim, as a separate preregistered stage per rung (section 6.8): on each of the four preregistered rungs $\mu \in \{0.1333, 0.1867, 0.2400, 0.4000\}$ independently, the certified-membership count of a Poisson sprinkling realized that rung’s independently certified continuum volume within $\tau = 2.5\%$, under a rule frozen before the data existed (CONCORDANT at every rung). For that Poisson-count stage we do not claim: any joint or composite cross-rung verdict, any statement at compactness between or beyond the tested rungs (no interpolation, no extrapolation), any promotion of the verdicts of sections 6–6.7 or of the auxiliary O4b audit, and nothing about causal-set theory or the discreteness of spacetime—a count agreeing with a volume says the instrument reproduces the measure it was pointed at, at one fixed diamond and one fixed density per rung. A literature priority search (August 2026) found partial overlap in four areas: causal-order-and-number/volume reconstruction (Braun 2025 [15]; Myrheim 1978 [8]), Schwarzschild sprinkle count verification (Homšak–Veroni [16] §V.1), analytic diamond-volume comparison theorems (Berthiere–Gibbons–Solodukhin [17]), and Schwarzschild causal-relation algorithms (He–Rideout [18]). The directed-

rounding volume enclosure and the preregistered TOST gate against it have no direct prior; the parallel Alfeyorov–Shnyukov CJ estimator [19] measures Weyl curvature from Hasse diagrams and is orthogonal. The equivalence gate applies established methodology—a confidence-interval form of the two one-sided tests procedure [20, 21] on exact Poisson (Garwood) [14] and binomial (Clopper–Pearson) [12] intervals, and equivalence-testing-based model validation [22]—with the frozen rule requiring the identified discrepancy, against the certified enclosure, of a conservative outer Garwood count interval (its lower limit from the certain count, its upper limit from the count including its ambiguous members, rescaled to a volume) to fall within the band (each Garwood tail at level 0.025, more conservative than the conventional 0.05). Its statistical novelty is thus the preregistered application against a certified enclosure, not the apparatus.

9. Limitations and future work

Results are controlled and mostly (1+1)D (dimension is checked to 4D, and the section 6 capstone is (3+1)D). The constants are convention-dependent (chain endpoint convention, null-inclusive causal relation, Myrheim–Meyer normalization); we state each where it is used. The spacelike proxy is exploratory. R5’s local-measure evidence is one profile family at one amplitude; the dose-response of the unweighted error floor to the profile amplitude is not measured. The capstone is bounded by its construction: Petrov type N with an exactly flat volume form—the property that makes the experiment clean also makes it unrepresentative of generic $\text{Weyl} \neq 0$ spacetimes—and by its single frozen operating point; generalizing beyond type N is now partially opened rather than closed: the Schwarzschild C1-paired extension is confirmed and the single-poset (C2-class) Schwarzschild stage is preregistered and detected with incomplete separation (section 6.7), and the diamond-volume oracle is now fully certified—the static-spacetime reductions derived in the public note (`docs/theory/schwarzschild_volume_oracle_note.md`) were completed by an MPFR directed-rounding flight-time certification, a uniform anchor-diamond containment proof, and a certified cell-refinement integrator, and the frozen configuration’s volume was computed once from an exact freeze checkout: $V \in [56.212737, 57.348019]$, relative half-width $0.009997 \leq 0.01$, `target-met` (`docs/prereg/p14_o3_volume.json`). That certification (O3) is the one the O4b audit below consumes. The section 6.8 count stage needed a narrower enclosure than its own tolerance could absorb, so the same configuration was re-certified at half the target width (O3’), giving $V \in [56.492959, 57.060667]$ —strictly inside O3, and the interval the $\mu = 0.1333$ rung is gated against (`docs/prereg/p14_o3p_volume.json`). The two are distinct certifications of the same diamond at different target widths, never combined.

An auxiliary instrument audit (O4b) then consumed that certified volume: at the single frozen Schwarzschild configuration, the sampler, the S1 volume response, and the certified oracle were found CONCORDANT under a frozen composite error budget of 3.25% with simultaneous coverage $\geq 95\%$ —G1 volume interval $[56.448806185841875, 56.9822829864225]$ against the certified $[56.212737, 57.348019]$ (identified discrepancy $[-0.8992123405928396, 0.7695461186219887]$, band 1.7034113309135284), G2 leak upper bound 0.14195058753928652 within budget 0.14195094424279403 with zero leaking points. The audit’s campaign run completed both gates and stopped at a publication-wiring defect; the verdict was recovered by re-applying the frozen decision functions to the preserved sufficient statistics (`run_kind: recovered_completed_campaign` in

`docs/prereg/p14_o4b_results.json`—no new seed, no solver call, no resampling, no gate change), with the recovery authenticated and bit-exactness enforced by contract tests (appendix B). This audit upgrades nothing in sections 6–6.7: it is a statement that the instrument stack agrees with the certified continuum volume at one frozen configuration—not a Poisson causal-set count verification, not mass- or domain-generality, not a C1/C2 joint verdict, and not complete separation or general volume accuracy. The prediction-anchored Poisson-count stage (sprinkle counts against ρV) is a separate stage, and it has since been executed across the four-rung mass ladder of section 6.8—CONCORDANT at $\mu \in \{0.1333, 0.1867, 0.2400, 0.4000\}$ —which is what carries the prediction-anchored claim; the O4b audit itself remains only the instrument statement described above. That ladder has its own bounds: four rungs are four points, the tolerance was chosen to be feasible rather than sharp, and the certification window itself closes at $M = 10/3$, where the photon sphere reaches the shell floor—the executed ladder stops one tenth of the shell floor short of that cliff, by certification rather than by taste. The natural next question—whether observer-relative distance *order* can be validated as recovering latent geometry, as opposed to being reconstructed from a supplied one—is the subject of a companion study that builds a preregistered discriminator on this foundation, measures its dose-response to geometry dilution and its dimension selection in (2+1)D, and then carries it to orders produced by growth dynamics and by an action-weighted 2D-order ensemble. At $N = 600$, random-start post-burn-in configurations at $\beta = 2$ and $\beta = 8$ pass the frozen instrument, while the $\beta = 32$ bipartite-start control blocks structurally. These are not certified equilibrium draws, and the companion study did not establish an equilibrium transition or finite-size scaling.

10. Reproducibility

Foundation-layer baseline commit 325df55. Every number in sections 4 and 5 has a cited `experiments/exp*.py` producer and expected summary path. All 19 cited summary tables are committed under `figures/data/`, regenerated in one recorded environment (CPython 3.11.9, numpy 2.4.6, matplotlib 3.11.1) from the unmodified baseline producers; `artifact_manifest.md` records the regeneration, per-file row counts, and raw SHA-256 digests under the repository’s LF storage-boundary rule, and a contract test recomputes every digest. The submission gate—the 19 legacy tables plus the finalized section 6 evidence bundle, provenance-locked—is met. Conventions (sprinkling measure, causal relation, chain and interval normalizations, Myrheim–Meyer inversion) are fixed in the foundation modules and stated in section 3.

Section 6 carries no regeneration debt: every number it cites lives in a committed artifact, and the repository’s test suite recomputes the capstone’s metrics, verdicts, and sentences from the stored raw samples, reproduces a prefix of every frozen seed stream, and asserts the commit-ancestry contract of appendix B. The stage runner (`experiments/positive_control/p14_prereg.py`) exposes three modes—`preflight`, `manifest`, `campaign`—and the campaign mode refuses to run unless the freeze manifest exists, its recorded preflight digest matches the committed artifact, and the certified source digests match the checked-out files.

Section 6.7 inherits the same standard for both of its stages: each runner (`experiments/positive_control/s4_schwarzschild_c1.py`, `experiments/positive_control/s5_schwarzschild_c2.py`) refuses to run unless an 8-file content-

addressed freeze manifest matches the checked-out protocol surface byte-for-byte, verified at entry and re-verified at exit, behind a fail-closed CLI; contract tests recompute the CONFIRMED (S4) and DETECTED (S5) verdicts—including the DeLong interval, the balanced-accuracy threshold and Clopper–Pearson–Bonferroni interval, and the frozen sentences—from the stored per-reading arrays through the frozen gate functions; and the seed ledger separates fresh allocation from deterministic replay, with replay output owned by a separate path that can never replace a fresh-observation artifact.

The auxiliary O4b instrument audit (section 9) meets the same standard with one addition. Its campaign runner refuses to run except on the clean exact checkout of the approved freeze SHA against a 25-file content-addressed manifest with an environment lock; the executed surface is preserved byte-for-byte as `docs/prereg/p14_o4b_executed_freeze_manifest.json`. Because the run stopped at a publication-wiring defect after both gates completed, the published verdict is a *recovery*: `experiments/oracle/o4b_recover.py` re-applies the frozen decision functions (empirical-Bernstein interval, Clopper–Pearson bound, frozen sizing constants) to the preserved sufficient statistics, authenticates its inputs against the SHA-256 of the committed incident and checkpoint records with a two-file cross-check, and refuses to publish unless the recomputation reproduces the preserved verdict bit-for-bit; contract tests (`tests/test_o4b_recover.py`) pin the recomputation, the tamper refusals, and the published artifact, and regression tests (`tests/test_o4b_wiring_fixes.py`) drive the repaired `main()` path end to end. No new seed was drawn, no solver called, no point resampled, and no gate changed anywhere in the recovery.

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How the output was verified, and how far that verification reaches. Every quantitative result in sections 4–6.8 is produced by a committed experiment script whose output file is pinned by a content-addressed manifest that a contract test recomputes, so the *artifacts* cannot drift unnoticed. The manuscript’s own transcription of those artifacts is machine-checked across the whole results arc: every figure printed in sections 4 through 6.7 is re-derived from its artifact and matched against the sentence that prints it, section 6.8’s per-rung table is re-derived the same way, and the auxiliary audit’s figures are pinned verbatim—so a hand-typed value fails the build, and so does one that silently moves to a neighbouring claim. The claim list is itself guarded: every numeral in sections 4–6.7 must be one the contract derives or one it names as an accepted exclusion with a reason, the exclusions being row selectors, rung labels and construction settings rather than results, so a figure added later cannot slip past unchecked. Provenance runs by two routes, both digest-locked at the artifact. The legacy results of sections 4 and 5 name their producing experiment (`exp03–exp23`) and its committed summary table, audited in `producer_audit.md`. The preregistered results of sections 6–6.8 name their own frozen artifacts—`p14_prereg_results.json` for

Table A1. Conventions and normalizations fixed in the foundation modules.

Quantity	Convention used
Causal relation	null-inclusive J^+ : i precedes j iff $t_j - t_i > 0$ and $(dt)^2 - (dx)^2 \geq -\text{atol}$, $\text{atol} = 1 \times 10^{-12}$; strict in time (irreflexive)
Longest chain	endpoint-exclusive count (interior elements, two fewer than the count including both endpoints); (1+1)D normalization $L \sim \sqrt{2\rho}\tau$ (asymptotic normalized value $\sqrt{2} \approx 1.414$), finite-size corrected
Interval volume, (1+1)D	open Alexandrov count K ; $V = K/\rho$; $V = \tau^2/2$, so $\tau_{\text{est}} = \sqrt{2K/\rho}$
Myrheim–Meyer	ordering fraction $r = (\text{related ordered pairs})/(N(N-1))$ inverted against $f(d) = \Gamma(d+1)\Gamma(d/2)/(4\Gamma(3d/2))$; equals the standard unordered form up to a factor 2 on both sides
Radar coordinates	τ_- = latest preceding tick, τ_+ = earliest succeeding tick; radar time $= (\tau_+ + \tau_-)/2$, radar distance $= (\tau_+ - \tau_-)/2$
Density (finite- N)	fixed- N sprinkle with empirical density; Poisson and fixed- N binomial sampling models both reported

the capstone, `p14_s4_results.json` and `p14_s5_results.json` for the Schwarzschild extensions, the per-rung count records for section 6.8, and `p14_o4b_results.json` for the auxiliary audit—each with its executed freeze manifest. What the contract does not do is stated too: it holds the printed characters to the committed artifact, not the artifact to nature, it re-runs no experiment, and it reaches sections 4–6.8 and the auxiliary audit’s figures only; numbers quoted elsewhere in the paper are still checked by reading. Writing it corrected two transcriptions that reading had passed (section 4.3’s radar-distance RMSE and section 6.4’s C2 separation statistic), which is why the reach is now stated as a contract rather than as care; the claim boundary’s copies of the same figures are held to the same artifacts, because that is where one of the two had been copied as well. The preregistered stages ran once each from an exact clean checkout of a merged freeze commit, under frozen decision rules, with their verdicts computed by the frozen code rather than asserted in prose. Every bibliographic entry was checked against an authoritative record (arXiv, Crossref, or the publisher) and then re-checked independently; the priority assessments in the companion records name the sources that were fetched.

Data availability statement

All data supporting the results of this study are version-controlled in the public repository at <https://github.com/lpaiu-cs/causal-spacetime>: the 19 digest-locked summary tables under `figures/data/`, and the preregistered evidence bundles and frozen artifacts of sections 6–6.8 and the auxiliary audit. Every cited file is pinned by a content-addressed manifest whose digests the repository’s contract tests recompute.

Appendix A. Conventions and normalizations

The conventions in table A1 are stated so the reported constants (e.g. the $\sqrt{2}$ chain normalization, the interval-volume factor) are unambiguous; each is fixed in the foundation modules.

Appendix B. Capstone execution provenance

The section 6 result is reproducible not merely in the sense that its code is committed, but in the sense that the *order of operations*—freeze before execution, execution before recording—is provable from the repository’s commit graph and enforced by mechanical gates.

Commit chain. Design and probe chain: PR #38–#48. Preregistration: preparation commit (PR #49), S1 cost measurement (PR #50), then $P' = 51875a2$ (a one-source seed-ledger fix) → preflight executed on a clean checkout of P' , its artifact recording `code_version` = P' and the SHA-256 digests of every execution-relevant source file → final-freeze commit $F = b858b08$ carrying the certification artifact and the freeze manifest (preflight digest, `preflight_code_version` = P' , the S1 price sentence, the frozen sizes and seeds; PR #51) → the campaign executed on a checkout of exactly F (not a descendant merge commit, so the recorded executing HEAD is F itself) → results commit $R = 5126ddf$, a direct child of F (PR #52). A test asserts $P' \prec F \prec R$ with `git merge-base --is-ancestor`.

Gates. Stochastic modes refuse a dirty worktree at entry. The campaign mode refuses to run unless (i) the freeze manifest exists, (ii) its recorded preflight digest matches the committed preflight artifact, and (iii) the source digests certified at preflight entry match the checked-out files—captured at entry and re-checked at exit, so a mid-run edit aborts the run with nothing recorded. Interrupted runs restart the same seed; an ambiguity violation blocks the stage rather than permitting a seed swap.

Artifacts. The complete section 6 evidence bundle, with digests, is enumerated in `artifact_manifest.md`; the per-sentence mapping from manuscript text to evidence file is recorded there as the citation-to-artifact inventory.

Frozen sentences. Each preregistered stage fixes, before execution, the sentence its verdict would license, and the campaign records that sentence in its results artifact. Those originals are written in Korean, the working language of this program’s records; every frozen sentence printed in section 6.4 and section 6.7 is therefore a translation, and the artifact holds the authoritative text. What keeps the two from drifting apart is not editorial care: a contract test reads each original from its artifact, holds it against the rendering the manuscript prints in its place, and fails if either side moves—so a translation cannot quietly soften, strengthen, or outrun the sentence the campaign actually licensed. The paper prints no Korean, and a separate test enforces that, because a script the reader cannot check would put the load-bearing wording beyond the reader’s reach.

Section 6.7 (Schwarzschild extension) provenance. Freeze commit `ceed85d` (PR #57, merge `5cffd90`) carries the frozen rule, the fresh seed registration, the runner, the contract tests, and the 8-file content-addressed freeze manifest—and no results. The campaign executed once on a clean checkout of exactly `ceed85d`; the runner captured the git state at entry, refused a dirty tree, verified every manifest digest at entry, and re-verified both at exit before writing, so the artifact’s recorded lineage is `start == end == ceed85d`, `clean`, `run_kind = fresh_observation`. The results commit (PR #58, merge `492e61f`) added the artifact, moved the campaign seed from fresh to observed in the ledger, and flipped the runner’s campaign path to a deterministic replay whose output is owned by a separate file and can never replace the fresh observation. Because that flip legitimately changed two protocol-surface files, the *current* tree’s manifest

hashes the post-result replay surface; the manifest that governed the executed run is preserved byte-for-byte as `docs/prereg/p14_s4_executed_freeze_manifest.json`, and a contract test verifies that snapshot against the historical blobs at `ceed85d` directly. Interrupted runs restart the same seed; undecided pairs are absorbed by identified intervals rather than a stop rule.

Section 6.7 second stage (S5) provenance. Freeze commit `1c26224` (PR #60, merge `3aeb8e5`) carried the rule, the fresh seed registrations, the runner, and the power certification; an execution review then caught an exact-versus-approximate chi-square mismatch in the certified resampling source, and the corrected freeze is `86e3674` (PR #61, merge `f286508`)—certification unchanged under the exact factor. The campaign executed once on a clean checkout of exactly `86e3674` (entry == exit git state, the 8-file content-addressed manifest verified at both ends, fail-closed CLI, `run_kind = fresh_observation`, curved seed `40000251` / flat seed `40000261`). The results commit (PR #62, merge `d1e0d20`) added the artifact, moved both seeds to observed, flipped the campaign path to a replay whose output is owned by a separate file, and preserved the manifest that governed the executed run byte-for-byte as `docs/prereg/p14_s5_executed_freeze_manifest.json`, verified against the historical blobs at `86e3674` by a contract test; the *current* `p14_s5_freeze_manifest.json` hashes the post-result replay surface.

Auxiliary O4b instrument-audit provenance (oracle arc). The certified oracle chain is $O1 \rightarrow O2 \rightarrow O3$ (PR #64–#68): flight-time certification, certified cell integrator, then the `O3` freeze (`785148e`) and its one approved execution producing the certified volume (`docs/prereg/p14_o3_volume.json`, executed surface `p14_o3_executed_freeze_manifest.json`). The first audit stage, `O4`, aborted at its `G3` with no verdict (PR #69–#70); its replay diagnostic and `G3` redesign reopened the preregistration (PR #71–#74), and `O4b` was frozen with new seeds (PR #75). The approved execution SHA is the freeze branch head `715865a`—not the merge commit. The campaign executed once from that clean exact checkout: `G3a` passed (71 wrapper-contract cases, 6,537 metered calls), `refs/o4b/reservation` was claimed (`46acee34`, retained permanently, spending seeds `40000401/40000411`), `G1` ran to 26,200,000 points and `G2` to 1,072,696 points, and the run stopped at the pre-publication provenance re-read on a wiring defect—the claim callback returned `None`—publishing nothing. The preservation commit (PR #76, merge `83e366c`) committed the incident and checkpoint verbatim (`p14_o4b_incident.json`, `p14_o4b_checkpoint.json`), retired both seeds in the ledger, and snapshotted the executed surface byte-for-byte (`p14_o4b_executed_freeze_manifest.json`, byte-identical to the freeze blob at `715865a`). The recovery commit (PR #77, merge `50586dd`) published `p14_o4b_results.json` with `run_kind = recovered_completed_campaign` by re-applying the frozen decision functions to the preserved statistics—bit-identical to the preserved verdicts, inputs authenticated by blob SHA-256 pins and an incident/checkpoint cross-check, the remote reservation re-verified immediately before publication; per the frozen stage-verdict table (`G1` and `G2` both concordant, `G3a` valid, frozen n reached) the stage verdict is `CONCORDANT`. The maintenance commit (PR #78, merge `a04e860`) fixed the wiring defects—the claim return value, the budget stage label, an explicit incident `failure_point`, a named checkpoint RNG stream—with end-to-end regression tests over the real `main()` path, leaving the result, incident, checkpoint, and executed snapshot untouched. No cam-

paign rerun and no same-seed re-entry occurred anywhere in this chain; the historical no-verdict records of the O4 abort and the O4b incident retain their original verdicts, with the recovery linked rather than retroactively regraded.

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